

IMPORTANCE OF

MANAGEMENT ACCOUNTING



PROVIDING DATA

The historical data gathered by managerial accounting demonstrates the growth of the business, which is useful for predicting.

ANALYZING DATA

Accounting data is presented understandably by calculating ratios and forecasting trends. This data is then examined for planning and decision-making purposes.



FACILITATE MEANINGFUL CONVERSATION

It helps in examining the status of the plans and the roles of the various departments in implementing them.

EMPHASIZES THE USE OF QUALITATIVE INFORMATION

It considers qualitative information that cannot be quantified in terms of money.



AIDS IN ACHIEVING GOALS

It helps in the facilitation of the conversion of organizational plans and objectives into attainable monetary objectives.

MBA FMA: Introduction to Management Accounting and Costing

Topic 6 A comprehensive guide to management accounting principles, cost classification, cost estimation, overhead allocation, and product and service pricing.

MBA FMA

TOPIC 6

Topic Overview

This topic covers the foundational concepts of management accounting and costing, structured across seven major sections:

01	02	03
6.1 Definition and Nature of MA What management accounting is and what it covers.	6.2 Role of Management Accounting Planning, decision-making, control, and strategy.	6.3 FAR vs MA Comparing financial and management accounting.
04	05	06
6.4 Cost Classification Seven major categories of cost classification.	6.5 Cost Estimation Engineering, judgment, high-low, and regression methods.	6.6 Overhead Allocation & Apportionment Tracing and sharing indirect costs.
07		
6.7 Costing and Pricing Products Goods purchased, manufactured, services, and combinations.		

6.1 Definition and Nature of Management Accounting

Management accounting (MA) is the process of **collecting, analyzing, interpreting, and presenting financial and non-financial information** to help managers plan, make decisions, control operations, and improve organizational performance.



What Management Accounting Covers

Revenues

Inflows of resources, mainly from the organization's core business operations.



Costs

Resources that have been foregone (outflows). Costs are further classified into:

- **Tangible costs** — can be measured in monetary terms
- **Intangible costs** — cannot be expressed in monetary terms

Management accounting focuses primarily on **tangible costs**, using revenue and cost information to aid management in making strategic decisions such as pricing, budgeting, make-or-buy (outsource), shutdown, product mix, and choosing among products.

- ❏ MA was previously called **cost accounting**, reflecting its original focus on recording and measuring costs of production.

6.2 Role of Management Accounting

Management accounting provides relevant financial and non-financial information to support a wide range of internal management functions:



Planning

Provides information for setting objectives, preparing budgets, and developing operational plans.



Decision-Making

Supplies relevant information to evaluate alternatives and make informed business decisions.



Cost Control

Identifies, analyses, and monitors costs to improve efficiency and reduce unnecessary expenditure.



Performance Evaluation

Measures performance of departments, managers, products, and business units against established targets.



Resource Allocation

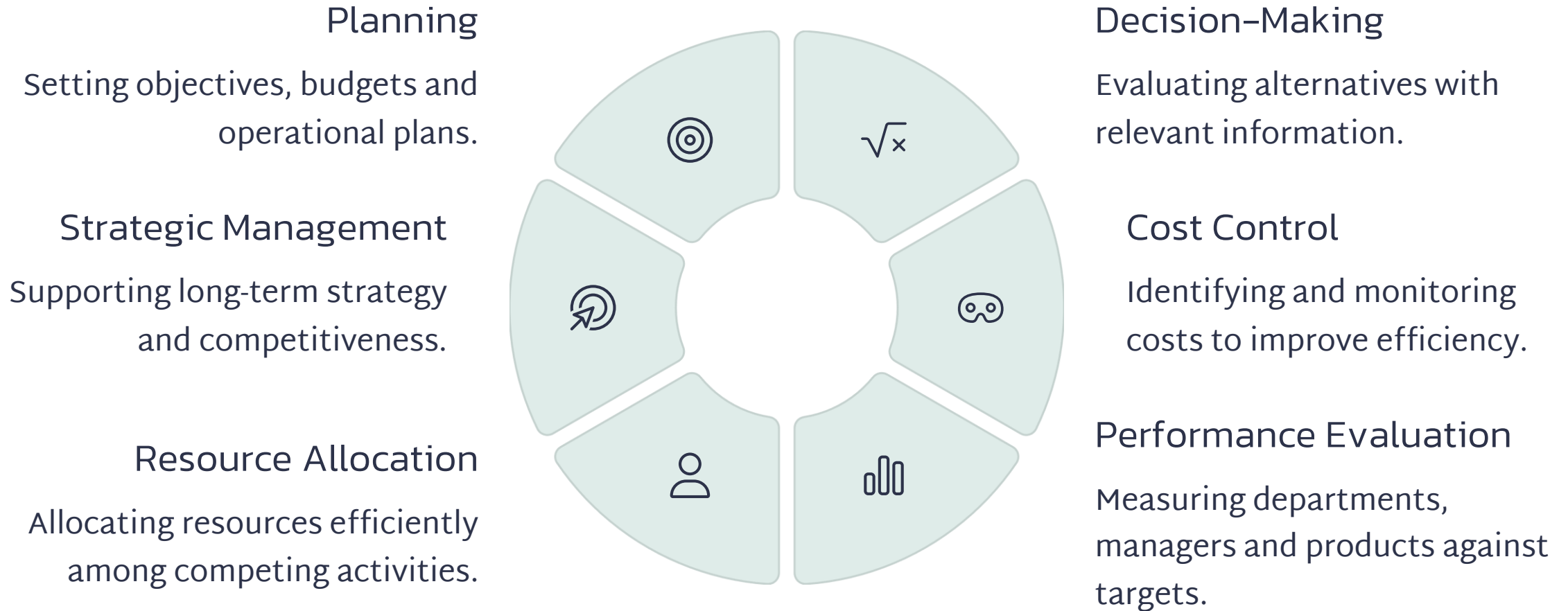
Helps management allocate financial and other resources efficiently among competing activities and projects.



Strategic Management

Provides financial and non-financial insights to support long-term strategy, competitiveness, and organizational sustainability.

Recap: Six Roles of Management Accounting



6.3 Financial Accounting vs Management Accounting

The table below compares the two disciplines across key dimensions:

Basis	Financial Accounting (FAR)	Management Accounting (MA)
Primary Purpose	Reports financial performance and position to external stakeholders.	Provides information to management for planning, decision-making and control.
Main Users	External: investors, lenders, regulators, tax authorities, shareholders.	Internal: managers, department heads, executives.
Focus	Historical financial transactions and events.	Both historical and future-oriented financial and non-financial information.
Reporting Requirements	Follows prescribed standards such as IFRS.	No mandatory standard format; designed according to management needs.
Time Orientation	Primarily past-oriented.	Past, present and future-oriented.
Level of Detail	Overall organizational level.	Detailed by product, department, branch, project or activity.
Frequency	Quarterly, semi-annually or annually.	Daily, weekly, monthly or whenever management requires it.
Type of Information	Mainly monetary and financial.	Both monetary and non-monetary, including operational data.
Confidentiality	Generally available to external stakeholders.	Usually confidential and intended for internal use only.
Key Emphasis	Accountability, compliance and financial reporting.	Planning, decision-making, performance management and control.

Recap: FAR vs MA — Key Differences

FAR



Users

External: investors, lenders, regulators.

Time Orientation

Past-oriented.



Standards

IFRS mandatory.

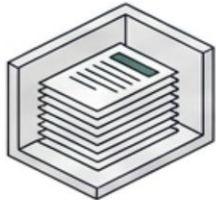
Frequency

Quarterly or annually.



Information Type

Monetary only.



Confidentiality

Publicly available.

MA

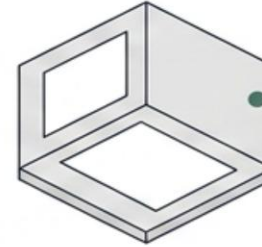
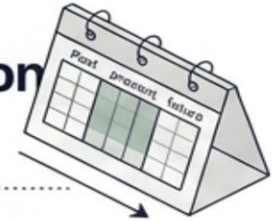


Users

Internal: managers, executives.

Time Orientation

Past, present, future.



Standards

No mandatory format.

Frequency

Daily, weekly, or as needed



Information Type

Monetary & non-monetary.



Confidentiality

Confidential internal use.



SECTION 6.4

Cost Classification

Cost classification is the process of grouping costs into meaningful categories based on their nature, behavior, purpose, or relationship with a product or activity, so that they can be properly analyzed and used by management.

6.4.1 Objectives of Cost Classification

1. Value Products

Classifies costs to determine the cost of producing a product or providing a service. Helps in inventory valuation, determining cost of sales, and setting appropriate selling prices.

2. Control Costs

Groups costs in a way that enables management to identify, monitor and compare costs against budgets or standards, thereby identifying inefficiencies and controlling unnecessary expenditure.

3. Make Decisions

Provides relevant cost information to help management evaluate alternatives and make decisions such as pricing, outsourcing, product mix, expansion, discontinuing a product, or accepting a special order.

6.4.2 Classification 1: Manufacturing vs Non-Manufacturing Costs

Manufacturing / Production Costs

Costs incurred in making or producing goods, including:

- **Direct Materials** — e.g., plastic and water to manufacture mineral water
- **Direct Labour** — wages of factory workers involved in direct production
- **Direct Costs** — e.g., direct taxes and royalties based on units manufactured
- **Production Overheads** — indirect costs supporting production: factory supervisor salary, depreciation, electricity, water, insurance

Non-Manufacturing / Non-Production Costs

Costs incurred outside the production process, mainly for administration, selling and distribution.

- Administration expenses
- Marketing and selling expenses
- Finance costs
- Research and other operational expenses not incurred in the factory

 The key question: **Where is the cost incurred?**

6.4.2 Classification 2: Product vs Period Costs

Product Costs

Costs attached to products and initially recorded as inventory costs; they become expenses **when the products are sold**.

Examples: direct materials, direct labour and factory overhead in manufacturing a table.

Period Costs

Costs charged as expenses in the accounting period in which they are incurred, rather than being included in product cost. Examples: office rent, administrative salaries and advertising expenses.



The key question: **Is the cost attached to the product or charged to the period?**

6.4.2 Classification 3: Cost Behaviour



Variable Costs

Total cost changes in direct proportion to the level of activity; cost per unit remains constant.

Example: material costs of Sh 500/unit — 100 units = Sh 50,000; 200 units = Sh 100,000.

Fixed Costs

Remain constant in total within a relevant range, regardless of changes in activity.

Example: factory rent of Sh 300,000/month whether 1,000 or 1,500 units are produced.

Semi-Variable / Mixed Costs

Contain both fixed and variable components; total cost changes with activity but not in direct proportion from zero.

Example: electricity with a fixed monthly charge plus a charge per unit consumed.

6.4.2 Classification 4: Relevance

Relevant Costs

Future costs that **differ between available alternatives** and therefore affect a decision. Example: if producing an extra 1,000 units requires Sh 200,000 of additional materials, the Sh 200,000 is relevant to the production decision.

Irrelevant Costs

Costs that **will not change** as a result of the decision and therefore should not influence it. Example: a machine's historical purchase cost (a sunk cost) when deciding whether to continue using or replace it.

❏ The key question: **Will the cost change because of the decision?**

6.4.2 Classifications 5, 6 & 7: Controllability, Avoidance & Discretion

Controllable vs Non-Controllable

Controllable: Costs a manager can influence within a given period e.g., overtime, material wastage, factory supplies.

Non-controllable: Costs a manager cannot significantly influence e.g., head-office charges allocated to a branch.

Avoidable vs Non-Avoidable

Avoidable: Can be eliminated if an activity or product is discontinued e.g., salary of a product-line supervisor.

Non-avoidable: Continue even if an activity is discontinued e.g., factory building rent after discontinuing one product line.

Discretionary vs Non-Discretionary

Discretionary: Amount or timing can be changed at management's discretion e.g., advertising, training, R&D, staff welfare.

Non-discretionary: Cannot easily be changed due to contractual, legal or operational commitments e.g., lease payments, statutory fees.

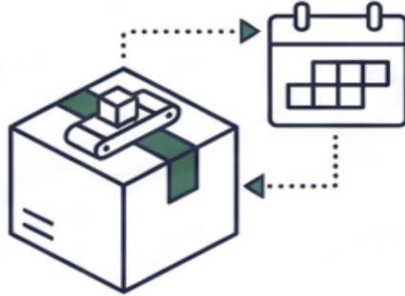
Cost Classification Summary

Manufacturing vs Non-manufacturing



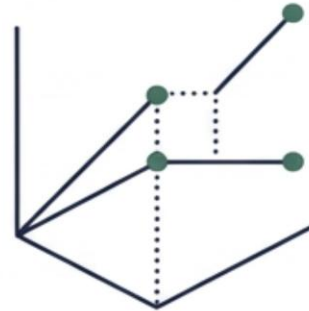
Where is the cost incurred?

Product vs Period



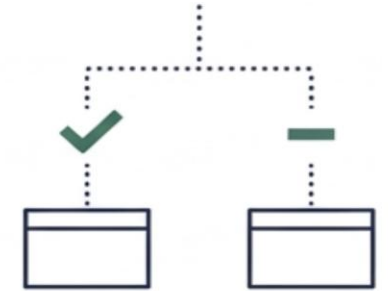
Is cost attached to product or charged to period?

Cost Behaviour



How does cost change when activity changes?

Relevant vs Irrelevant



Will cost change because of decision?

Controllable vs Non-controllable



Can manager influence cost?

Avoidable vs Non-avoidable



Can cost be eliminated by stopping activity?

Discretionary vs Non-discretionary



Does management have discretion over cost?

SECTION 6.5

Cost Estimation

Cost estimation is the process of predicting the amount of cost that an organization is likely to incur at a particular level of activity, using historical data, technical information, expert knowledge or statistical techniques. It supports budgeting, pricing, cost control and decision-making. Four methods are covered: **engineering method**, **professional judgment**, **high-low method**, and **regression method**.



6.5.1 Engineering Method

The engineering method estimates costs by analyzing the **physical inputs, production processes and technical requirements** needed to produce a product or provide a service. It is suitable where there is no historical data about costs of production. Costs are estimated by having several **production runs** a single, continuous batch of production during which a specified quantity of identical or similar products is manufactured using the same setup. Production runs help separate costs into variable and fixed components.

Benefits

- **High accuracy:** Uses detailed technical analysis of materials, labour, machine time and production processes to develop realistic estimates.
- **Useful for new products:** Can estimate costs even when historical cost data is unavailable or unreliable.

Limitations

- **Time-consuming and costly:** Requires detailed engineering studies and specialist expertise to analyse production processes.
- **Potentially subjective:** Estimates may depend on engineers' assumptions and judgements about resource usage and production efficiency.

6.5.2 Professional Judgment Method

The professional judgment method estimates costs using the **experience, knowledge and informed judgment** of managers, accountants, engineers or other specialists. It is particularly useful when launching a product that has data from other similar businesses. Experts assess available information and use their experience to determine expected fixed and variable costs.

Benefits

- **Simple and quick:** Uses managers' or experts' experience and knowledge to produce cost estimates without complex calculations.
- **Useful when data is limited:** Can be applied where historical cost data is unavailable, incomplete or unreliable.

Limitations

- **Subjective:** Estimates depend heavily on the individual's experience, assumptions and judgment, which may introduce bias.
- **Less precise:** May produce inconsistent or inaccurate estimates, particularly when the business environment or production conditions change significantly.

6.5.3 High-Low Method — Introduction

The high-low method is a simple technique for separating a mixed cost into its **variable and fixed components** by using the total cost and activity levels at the highest and lowest points. It is suitable when historical cost and activity data are available, the cost is approximately linear, and management needs a quick and simple estimate.

The cost function is: $Y = a + bX$, where:

Y
Total cost (Sh million)

a
Fixed cost (Sh million)

b
Variable cost per 1,000
units (Sh million)

X
Quantity produced
(‘000 units)

Historical data for the example:

Production Quantity (X) — ‘000 units	Total Cost (Y) — Sh million
2 — Low	10
3	13
4	16
5	19
6 — High	22

High-Low Method — Steps 1 to 3

1 Calculate Variable Cost (b)

Formula: $b = (\text{High cost} - \text{Low cost}) \div (\text{High activity} - \text{Low activity})$
 $b = (22 - 10) \div (6 - 2) = 12 \div 4 =$
Sh 3 million per 1,000 units

2 Calculate Fixed Cost (a)

Using the high activity level: $Y = a + bX$
 $22 = a + (3 \times 6) \rightarrow 22 = a + 18 \rightarrow \mathbf{a =}$
Sh 4 million

3 Determine the Cost Function

Substituting values: $\mathbf{Y = 4 + 3X}$
Fixed cost = Sh 4 million;
Variable cost = Sh 3 million per 1,000 units; X = production quantity ('000 units); Y = total cost (Sh million).

High-Low Method — Step 4: Estimate Cost at 5,000 Units

If the company plans to produce 5,000 units, then $X = 5$: $Y = 4 + (3 \times 5) = 4 + 15 = \text{Sh } 19 \text{ million}$

Component	Calculation	Cost (Sh million)
Fixed cost	—	4
Variable cost	3×5	15
Total cost	$4 + 15$	19

Conclusion: The High-Low method estimates total production cost as $Y = 4 + 3X$ — the company incurs Sh 4 million of fixed costs plus Sh 3 million for every additional 1,000 units produced.

Benefits

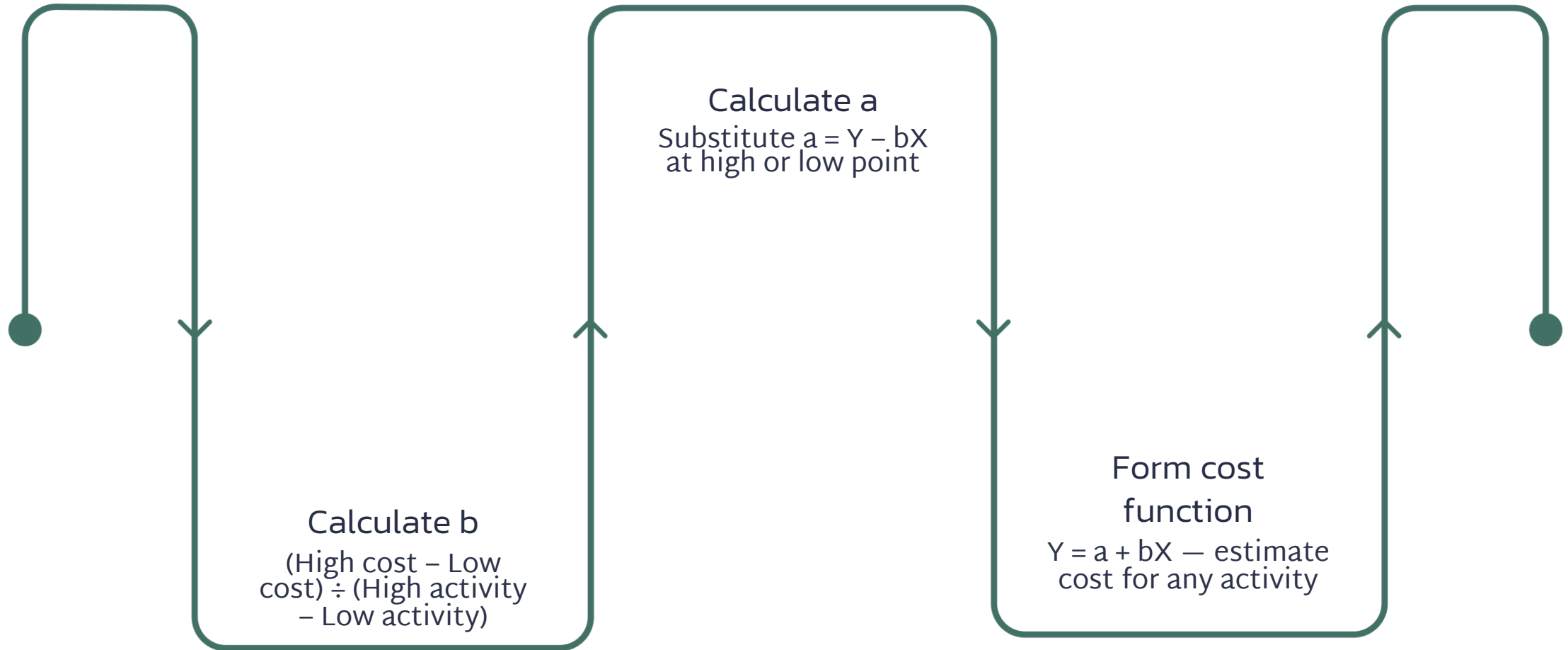
- **Simple and quick:** Requires only the highest and lowest activity observations.
- **Useful for short-term planning:** Provides a quick estimate for budgeting and decision-making.

Limitations

- **Uses only two observations:** Ignores all other available data, making the estimate less reliable.
- **Sensitive to extreme values:** Unusual high or low activity levels can distort the estimated variable and fixed costs.

Practice Focus: High-Low Method

Remember the three-step process for the High-Low method:



Always check: are the high and low points based on **activity level** (not cost)?
Abnormal observations should be excluded before applying the method.

6.5.4 The Regression Method — Introduction

Simple linear regression is a statistical technique used to estimate the relationship between total cost (Y) and a single activity variable (X) using historical observations. The basic cost function is: $Y = a + bX$, where:

Y

Estimated total
cost

a

Fixed cost (the
intercept)

b

Variable cost per
unit of activity (the
slope/coefficient)

X

Level of activity or
quantity produced

Regression estimates a and b by finding the **line of best fit** through the historical cost and activity data, minimizing the overall differences between actual and estimated costs.

Regression Method — Illustrative Output

A manufacturing company runs a simple linear regression with **Dependent variable (Y): Total Production Cost** and **Independent variable (X): Production Quantity** (X in '000 units; Y in Sh million).

Regression Statistics

Statistic	Output
Multiple R	0.990
R Square (R ²)	0.980
Adjusted R Square	0.977
Standard Error	0.650
Observations	10
F-statistic	392.00
Significance F	0.000001

Regression Coefficients

Variable	Coeff.	t-Stat	P-value
Intercept	4.20	7.64	0.00004
Prod. Qty (X)	2.80	19.80	0.000001

95% Confidence Intervals: Intercept = Sh 2.93–5.47 million; X coefficient = Sh 2.48–3.12 million per 1,000 units.

Regression Method — Cost Function and Estimation

From the regression output: $a = 4.20$ and $b = 2.80$. Therefore: $Y = 4.20 + 2.80X$

- **Fixed cost (a)** = Sh 4.20 million
- **Variable cost (b)** = Sh 2.80 million per additional 1,000 units produced

Suppose management plans to produce **8,000 units** ($X = 8$):

$$Y = 4.20 + (2.80 \times 8) = 4.20 + 22.40 = \text{Sh } 26.60 \text{ million}$$

Cost Component	Calculation	Sh million
Fixed cost	4.20	4.20
Variable cost	2.80×8	22.40
Total estimated cost	$4.20 + 22.40$	26.60

Interpreting Regression Statistics

Statistic	Result	Interpretation
R	0.990	Very strong positive relationship between production quantity and total cost.
R ²	0.980	~98% of the variation in total cost is explained by production quantity , very strong model fit.
Adjusted R ²	0.977	Model explains ~97.7% of cost variation after adjustment for degrees of freedom.
F-statistic	392.00	Tests whether the regression model as a whole provides a statistically significant explanation of the dependent variable.
Significance F	0.000001	Less than 0.05, the overall regression model is statistically significant at the 5% level.
Std. Error	0.650	Typical size of the regression prediction error, measured in Sh million.
t-statistic	7.64 / 19.80	Tests whether each individual coefficient is significantly different from zero.
P-value	0.00004 / 0.000001	Both below 0.05, intercept and production coefficient are statistically significant at the 5% level.
Lower 95%	2.93 / 2.48	Lower boundary of the 95% confidence interval for the respective coefficient.
Upper 95%	5.47 / 3.12	Upper boundary of the 95% confidence interval for the respective coefficient.

Confidence Intervals and Overall Interpretation

95% Confidence Intervals

Fixed cost (intercept): Sh 2.93 million to Sh 5.47 million, the underlying fixed-cost coefficient is estimated to lie within this range with 95% confidence.

Variable cost coefficient: Sh 2.48 million to Sh 3.12 million per 1,000 units the estimated variable cost is Sh 2.80 million, with a 95% confidence interval of Sh 2.48–3.12 million per additional 1,000 units.

Overall Interpretation

Production quantity is a **statistically significant and very strong predictor** of total production cost: high R^2 of 98%, statistically significant F-test, and a statistically significant production coefficient with a very small P-value.

The estimated cost function is: $Y = 4.20 + 2.80X$

At 8,000 units, estimated production cost = **Sh 26.60 million**



Statistical significance does not by itself prove the model is appropriate. Management should also consider economic reasonableness, data reliability, regression assumptions, and whether the estimated production level falls within the relevant range.

Why Regression is Superior to the High-Low Method

→ Uses All Observations

Regression uses the entire historical dataset; the High-Low method uses only the highest and lowest activity observations, making regression generally more reliable.

→ Best-Fit Line

Determines the line of best fit that minimizes overall differences between actual and estimated costs, rather than basing the estimate on just two points.

→ Measures Model Fit (R^2)

R^2 shows the proportion of variation in total cost explained by the activity variable; a high R^2 indicates a good fit to historical data.

→ Tests Overall Significance

The F-statistic and P-value assess whether the regression model as a whole provides a statistically significant explanation of cost behaviour.

→ Tests Individual Cost Drivers

The t-statistic and P-value indicate whether the estimated variable-cost coefficient is statistically significant, helping determine whether the selected activity driver is genuinely associated with cost.

→ Provides Confidence Intervals & Diagnostics

Regression provides 95% confidence intervals for coefficients, measures of estimation error, and enables diagnostic analysis of residuals, outliers and unusual observations.

Regression: Benefits and Limitations

Benefits

- **More reliable estimates:** Uses all available historical observations to calculate the best-fitting relationship between activity and cost.
- **Tests model reliability:** Provides R^2 , F-statistics, t-tests, P-values and confidence intervals to assess the strength, significance and suitability of the cost function.
- **Supports better forecasting:** Once statistically evaluated and found suitable, the estimated cost function $Y = a + bX$ can be used to forecast costs at different production levels with greater confidence.

Limitations

- **Requires reliable data and expertise:** Regression can be difficult to apply and interpret without appropriate statistical knowledge and quality historical data.
- **Depends on assumptions:** Results may be misleading if assumptions such as linearity, independence, constant variance and absence of influential outliers are violated.

Practice Focus: Regression Interpretation

When interpreting regression output, focus on these key statistics:

R² (Coefficient of Determination)

Proportion of variation in total cost explained by the activity variable. $R^2 = 0.98$ means 98% of cost variation is explained by production quantity.

Significance F and P-values

Both must be below 0.05 for the model and individual coefficients to be statistically significant at the 5% level.

Coefficients (a and b)

The intercept (a) = fixed cost; the slope (b) = variable cost per unit of activity. These form the cost function $Y = a + bX$.

95% Confidence Intervals

The range within which the true fixed and variable cost parameters are estimated to fall with 95% confidence.

Recap: Four Cost Estimation Methods

Engineering Method

Analyzes physical inputs and technical requirements. Best for new products with no historical data. Uses production runs to separate fixed and variable costs.

Professional Judgment

Uses expert experience and knowledge. Quick and simple. Best when launching a product with data from similar businesses. Subjective and potentially biased.

High-Low Method

Uses only the highest and lowest activity observations. Formula: $b = (\text{High cost} - \text{Low cost}) \div (\text{High activity} - \text{Low activity})$. Simple but sensitive to extreme values.

Regression Method

Statistical technique using all historical observations. Provides R^2 , F-statistic, t-tests, P-values and confidence intervals. More reliable but requires statistical expertise.

SECTION 6.6

Overhead Allocation and Apportionment

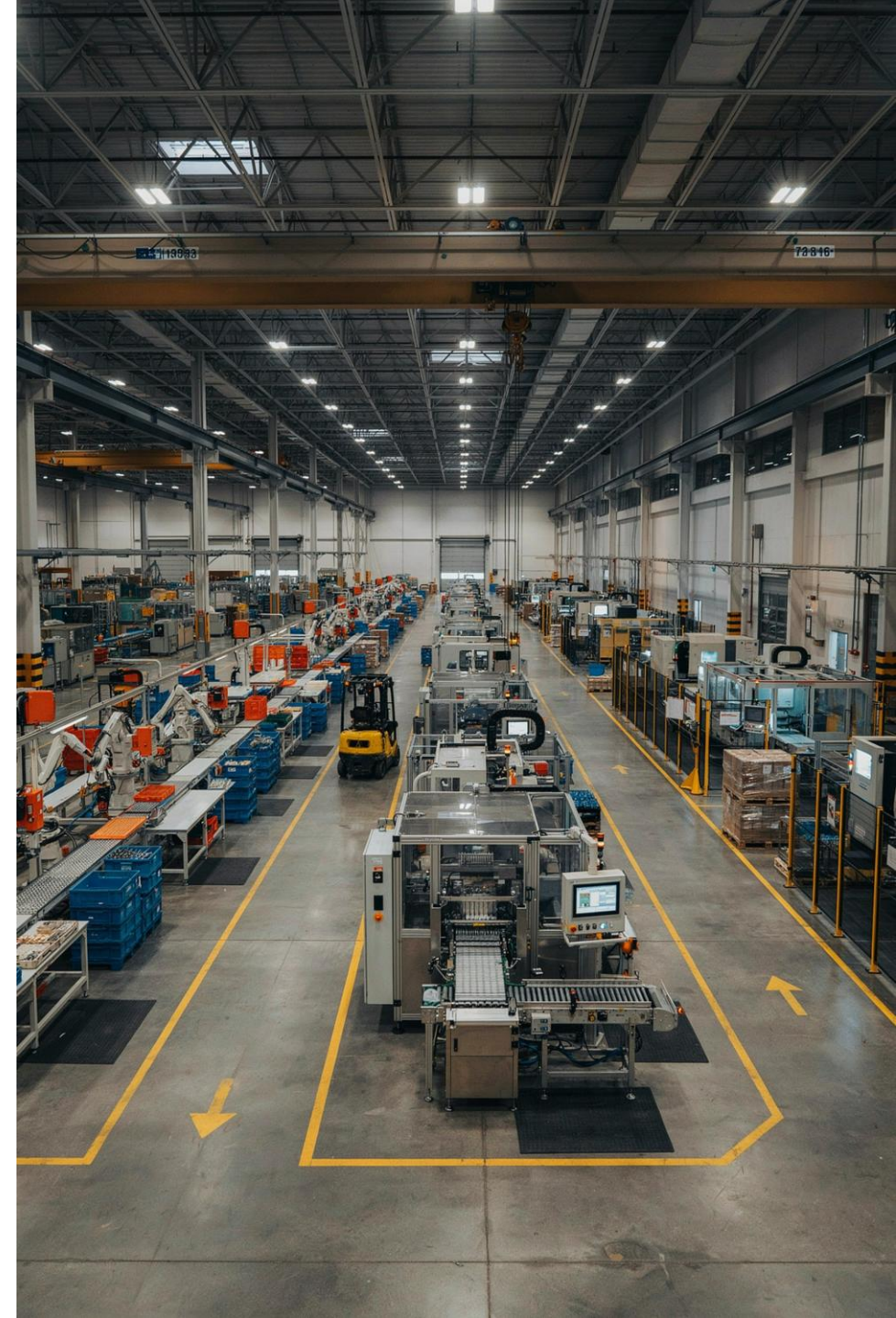
When an organization produces more than one product or provides several services, indirect costs (overheads) must be assigned to each product or service. There are two situations:

Overhead Allocation

If overheads can be **traced to a specific product**, they are directly included in the costing of that product.

Overhead Apportionment

If overheads **cannot be traced** to a specific product, they are shared among products using a suitable basis or ratio.



Bases of Overhead Apportionment

Type of Overhead	Reasonable Basis	Explanation / Example
Depreciation & insurance of assets	Value of assets / Net book value / Replacement value	A department with 40% of total asset value receives ~40% of the overhead.
Employee benefits & supervision	Number of employees / Labour hours / Direct labour cost	Employee welfare costs apportioned based on number of employees.
Electricity	Kilowatt-hours (kWh) consumed / Machine hours	Where meters are available, use actual consumption; otherwise machine hours.
Rent, rates & insurance of factory	Floor area occupied (square metres)	Costs allocated according to physical space occupied by each department.
Light and heat	Floor area / Cubic capacity / Number of employees	Floor area for lighting; cubic capacity may be more appropriate for heating.
Maintenance costs	Machine hours / Value of machinery / Maintenance hours	Allocated according to the extent each department uses or requires maintenance.

❏ The best basis of apportionment is the factor that most closely reflects the **cause** of the overhead cost e.g., floor area for rent, kWh for electricity, machine hours for maintenance.

Recap: Overhead Allocation vs Apportionment

Overhead Allocation

When overheads can be **directly traced** to a specific product, they are allocated directly to that product's cost.

Overhead Apportionment

When overheads **cannot be traced** to a specific product, they are shared using a suitable basis e.g., floor area for rent, kWh for electricity, machine hours for maintenance.

The best basis of apportionment is the factor that most closely reflects the **cause** of the overhead cost.



SECTION 6.7

Costing and Pricing Products

Once costs are known, the next step is to determine the cost of products and set their prices. Organizations may buy goods for resale, manufacture goods, provide services, or offer a combination of goods and services. Each situation requires a different costing approach.

01

6.7.1 Pricing Goods Purchased for Resale

IAS 2 cost of purchase plus markup or margin.

02

6.7.2 Pricing Goods Manufactured

Direct costs plus absorbed production overheads (traditional or ABC).

03

6.7.3 Pricing Services

Identifying a cost unit and accumulating direct and indirect costs.

04

6.7.4 Combination of Goods and Services

Cumulating all costs and using the good as the cost unit.

6.7.1 Pricing Goods Purchased for Resale — IAS 2

Under IAS 2, the cost of purchased finished goods and raw materials includes **all costs of purchase and other costs incurred in bringing the inventories to their present location and condition.**

Included in Inventory Cost

- Purchase price
- Import duties and non-recoverable taxes
- Transport to warehouse/point of sale
- Handling and directly attributable acquisition costs
- Less: trade discounts, rebates and similar deductions

Excluded from Inventory Cost

- Administrative overheads that do not contribute to bringing inventory to its present condition
- Selling costs
- Storage costs not necessary for the production process

Once total cost is determined, the price is computed by adding gross profit: **Cost + Gross Profit = Selling Price**

- **Markup** = Gross Profit as a % of Cost (Cost = 100%)
- **Gross Profit Margin** = Gross Profit as a % of Selling Price (Selling Price = 100%)

Practice Focus: IAS 2 — What to Include and Exclude

Always Include

- Purchase price / supplier invoice
- Import duties and non-recoverable taxes
- Freight, shipping and transport to warehouse
- Clearing and handling charges
- Direct materials, direct labour, direct expenses
- Production overheads (absorbed)
- Normal waste and normal idle time

Always Exclude

- Administrative overheads not contributing to bringing inventory to its present condition
- Selling and advertising costs
- Storage costs not necessary for the production process
- Abnormal waste and abnormal idle time
- Trade discounts and rebates (deducted from cost)

Markup vs Margin — Concept

Using the example: Cost = 80, Gross Profit = 20, Selling Price = 100:

Markup (on Cost)

$$\text{Gross Profit} \div \text{Cost} \times 100$$

$$= 20 \div 80 \times 100 = \mathbf{25\%}$$

Cost is 100%; Selling Price is 125%.

Gross Profit Margin (on Sales)

$$\text{Gross Profit} \div \text{Selling Price} \times 100$$

$$= 20 \div 100 \times 100 = \mathbf{20\%}$$

Selling Price is 100%; Cost is 80%.

- ❏ For margin calculations, work backwards: start with Selling Price = 100%, subtract the gross profit % to get the cost %, then use the law of proportions to compute Sh amounts.

IAS 2 Cost Example — Kenyan Trading Company

A Kenyan trading company imports 1,000 units of a finished product for resale. The following costs are incurred:

Cost Item	Total (Sh)	Per Unit (Sh)	Treatment
Supplier's invoice price	1,000,000	1,000	Included
Import duty	100,000	100	Included
Freight and shipping	150,000	150	Included
Clearing and handling charges	50,000	50	Included
Non-refundable import taxes	80,000	80	Included
Transport from port to warehouse	40,000	40	Included
Warehouse storage after receipt	30,000	30	Excluded
Head-office administration	50,000	50	Excluded
Selling and advertising costs	60,000	60	Excluded

Total inventory cost = Sh 1,000,000 + 100,000 + 150,000 + 50,000 + 80,000 + 40,000 = **Sh 1,420,000**

Cost per unit = Sh 1,420,000 ÷ 1,000 = **Sh 1,420**

Estimating Selling Price — 25% Markup vs 25% Margin

Item	Markup %	Markup Sh	Margin %	Margin Sh	Note
Cost	100%	1,420	75%	1,420	
Gross Profit	25%	355	25%	473	
Selling Price	125%	1,775	100%	1,893	

Under **markup**: if 100% = Sh 1,420, then 25% = $(25/100 \times 1,420) = \text{Sh } 355$; Selling Price = $(125/100 \times 1,420) = \text{Sh } 1,775$.

Under **margin**: Selling Price = 100%, Cost = 75%. Therefore Selling Price = $(100/75 \times 1,420) = \text{Sh } 1,893$.

6.7.2 Pricing Goods Manufactured — IAS 2

Under IAS 2, the cost of goods manufactured includes the costs of production. The following are **included** and **excluded**:

Included in Manufacturing Cost

- Direct materials (less abnormal waste)
- Direct labour (including normal idle time)
- Direct expenses
- Production overheads
- Normal waste/loss (expected material losses and normal idle labour)

Excluded from Manufacturing Cost

- Abnormal waste (excess loss in direct materials or excess idle labour)
- Storage costs not necessary for the production process
- Administrative overheads not contributing to bringing inventory to its present condition
- Selling costs

Production overheads (e.g., water, electricity) are not known in advance, so the concept of **overhead absorption** is used. Two approaches exist: the **traditional method** and **activity-based costing (ABC)**.

Traditional Overhead Absorption

Under the traditional method, overhead absorption is the process of allocating production overhead costs to units of production using a **predetermined absorption rate**:

Absorption Rate = Budgeted Production Overheads ÷ Budgeted Absorption Base
(e.g., direct labour hours, machine hours or units produced)

The resulting absorbed overhead is included in the product cost for inventory valuation, pricing and profitability analysis.

A manufacturing company produces a batch of 1,000 finished units. The company uses **material cost** and **direct labour hours** as bases for absorbing production overheads.

Manufacturing Example — Direct Materials

The production process uses three direct materials, each subject to normal production losses. Abnormal losses are excluded from product cost.

Material	Qty Purchased	Cost/Unit	Total Cost (KSh)	Normal Loss	Actual Loss	Abnormal Loss
Material A	1,050 kg	200	210,000	5% = 52.5 kg	72.5 kg	20 kg
Material B	500 kg	300	150,000	4% = 20 kg	25 kg	5 kg
Material C	300 kg	300	90,000	2% = 6 kg	6 kg	Nil
Total			450,000			

Manufacturing Example — Direct Labour and Other Direct Costs

The batch passes through three production processes. The 60 hours of idle time are **normal idle time** arising from unavoidable production interruptions and are included in product cost.

Process	Productive Hours	Idle Hours	Total Paid Hours	Rate/Productive Hour	Direct Labour Cost
Process 1	380	20	400	KSh 500	KSh 190,000
Process 2	270	30	300	KSh 600	KSh 162,000
Process 3	240	10	250	KSh 700	KSh 168,000
Total	890	60	950		KSh 520,000

Other Direct Cost: Excise tax of Sh 80,000 directly attributable to the production of the batch (non-refundable).

Production Overheads: Total Sh 300,000 — Sh 150,000 absorbed on direct material cost (rate = 30% × material cost) and Sh 150,000 absorbed on direct labour hours (rate = Sh 187.25 per direct labour hour).

Manufacturing Example — Full Cost Solution

Cost Item	Computation	Final Cost (Sh)
Material A	$210,000 - (20 \text{ kg} \times 200)$	206,000
Material B	$150,000 - (5 \text{ kg} \times 300)$	148,500
Material C	No abnormal loss	90,000
Total Direct Materials	$206,000 + 148,500 + 90,000$	444,500
Total Direct Labour	$190,000 + 162,000 + 168,000$	520,000
Other Direct Cost (Excise Tax)	Directly attributable, non-refundable	80,000
Production Overhead — Materials	$30\% \times 444,500$	133,350
Production Overhead — Labour	$890 \text{ hrs} \times 187.25$	166,652.50
Total Cost of Batch	$444,500 + 520,000 + 80,000 + 300,002.50$	1,344,502.50
Cost per Unit	$1,344,502.50 \div 1,000$	Sh 1,344.50
Markup — 30%	$1,344.50 \times 30\%$	Sh 403.35
Selling Price per Unit	$1,344.50 + 403.35$	Sh 1,747.85 $\approx$ Sh 1,750

Activity-Based Costing (ABC) — Introduction

When an organization produces several products, it can use either the **traditional approach** or **activity-based costing (ABC)** for production overhead absorption. Under ABC, overheads are assigned based on the **activities actually consumed** by each product, rather than a single volume-based absorption rate. This provides a more accurate picture of the true cost of each product.



ABC — Absorption Rates

Four activity cost pools are identified, each with its own cost driver:

Activity	Total Overhead	Cost Driver	Rate
Machine Processing	Sh 300,000	3,000 machine hours	Sh 100 per machine hour
Machine Setup	Sh 150,000	30 setups	Sh 5,000 per setup
Quality Inspection	Sh 100,000	50 inspections	Sh 2,000 per inspection
Material Handling	Sh 100,000	80 movements	Sh 1,250 per movement

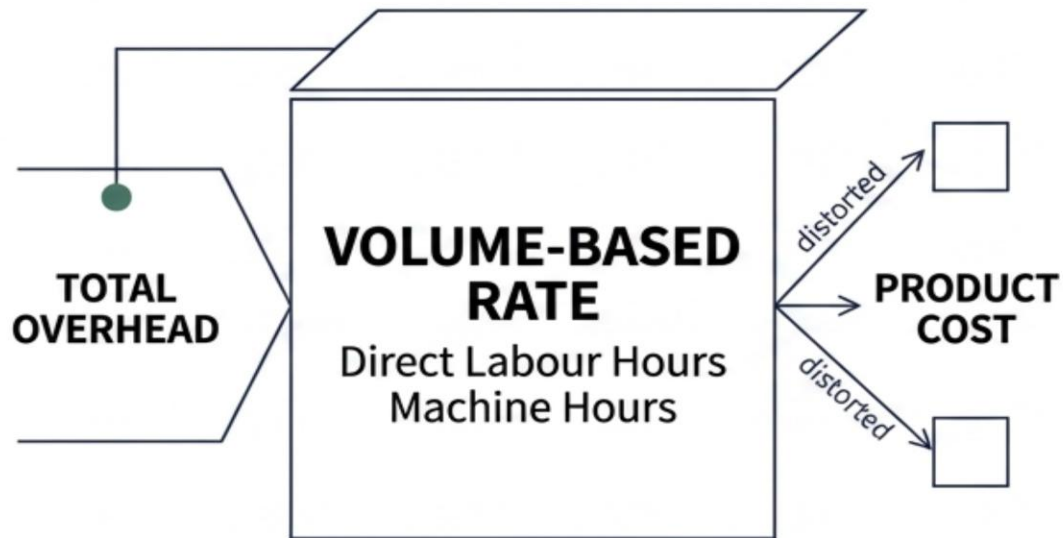
ABC — Product Costing Solution

Costing Item	Product A Computation	A (Sh)	Product B Computation	B (Sh)
Direct materials	1,000 × Sh 600	600,000	500 × Sh 900	450,000
Direct labour	1,000 × Sh 400	400,000	500 × Sh 600	300,000
Machine processing	2,000 hrs × 100	200,000	1,000 hrs × 100	100,000
Machine setup	10 setups × 5,000	50,000	20 setups × 5,000	100,000
Quality inspection	20 × 2,000	40,000	30 × 2,000	60,000
Material handling	30 × 1,250	37,500	50 × 1,250	62,500
Total Production Cost	600,000 + 400,000 + 327,500	1,327,500	450,000 + 300,000 + 322,500	1,072,500
Units produced		1,000		500
Cost per unit	1,327,500 ÷ 1,000	1,327.50	1,072,500 ÷ 500	2,145.00
Gross profit margin	Given	25%	Given	25%
Selling price per unit	1,327.50 ÷ (1 – 25%)	1,770.00	2,145 ÷ (1 – 25%)	2,860.00

- ❑ ABC assigns overhead based on activities actually consumed by each product. Product B absorbs significantly more overhead per unit despite having a lower total overhead allocation than Product A, because it requires more setups, inspections and material movements.

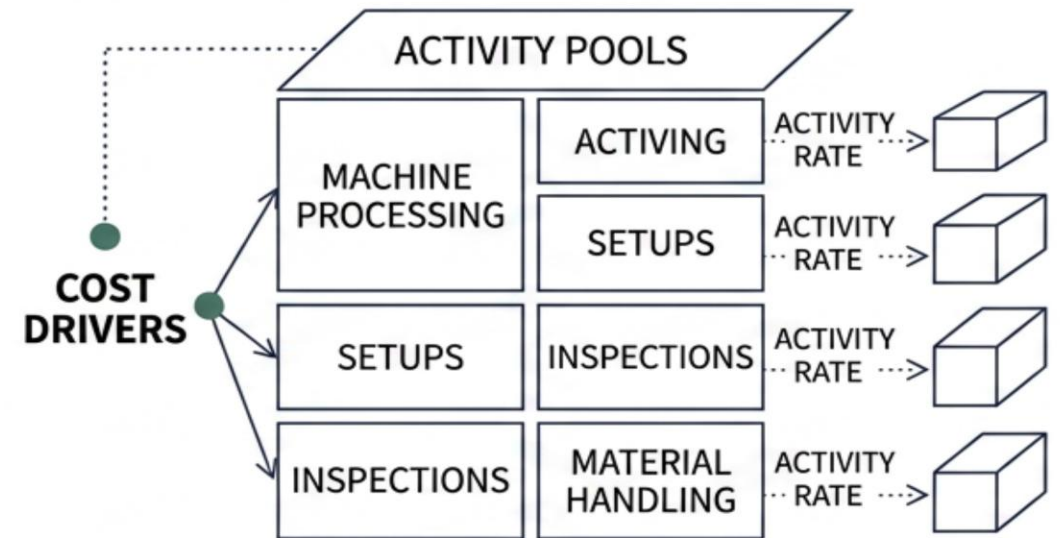
Practice Focus: Traditional vs ABC Overhead Absorption

TRADITIONAL OVERHEAD ABSORPTION



Single volume-based rate.
Simple, but may distort costs
for complex products.

ACTIVITY-BASED COSTING (ABC)



Multiple activity-based rates.
Accurately assigns overhead
based on activities consumed.



6.7.3 Pricing Services — Introduction

Costing services is more challenging than costing goods because services are **intangible** and cannot be separately identified. The first step is to identify a suitable basis for measuring costs, called the **cost unit**. The cost unit depends on the type of service or industry.

Cost Units by Service Sector

Service Sector	Typical Cost Unit	What Is Being Costed
Hospital / Healthcare	Patient-day / Bed occupied	Cost of treating one patient for one day
Hotel	Room-night	Cost of providing one occupied room for one night
Transport – Bus / Airline	Passenger-kilometre	Cost of transporting one passenger for one kilometre
Freight / Logistics	Tonne-kilometre	Cost of transporting one tonne of goods for one kilometre
School / University	Student-hour / Student-year	Cost of providing education to one student for a year
Banking	Transaction / Account maintained	Cost of processing one banking transaction
Legal / Accounting	Billable hour / Assignment	Cost of providing one professional hour or completing an assignment
Telecommunications	Call-minute / Data GB	Cost of providing one minute of calls or one GB of data
Security services	Guard-hour / Guard-day	Cost of providing one security guard for one hour or day
Construction services	Labour-hour / Square metre	Cost of providing construction work per labour hour or square metre

Service Costing Example — Two-Year MBA Programme

A university costs a two-year MBA programme for one student, comprising **16 taught units** (8 per year) and **1 research thesis**. The university adds a **20% markup** on total cost to determine the final fee. All amounts in USD.

Direct Costs per Student

Direct Cost Item	Year 1 (USD)	Year 2 (USD)	Total (USD)
Taught units — lecturer costs (8 × \$400)	3,200	3,200	6,400
Learning materials (16 × \$75)	600	600	1,200
Assessments/examinations (16 × \$50)	400	400	800
Thesis supervision	—	1,200	1,200
Thesis examination	—	500	500
Total direct costs	4,200	5,900	10,100

MBA Programme — Indirect Costs and Total Cost

Indirect Costs per Student

Indirect Cost Item	Basis of Allocation	Cost per Student (USD)
University administration	Allocated admin cost per student	1,000
Library and electronic databases	Annual cost per student × 2 years	800
ICT and learning management system	\$300 per year × 2 years	600
Classrooms and facilities	Allocated facilities cost	1,000
Student support services	Allocated student-service cost	400
Utilities and campus services	Allocated utilities cost	500
Marketing and admissions	Allocated cost per student	300
Total indirect costs		4,600

Total cost of educating one MBA student: Direct costs \$10,100 + Indirect costs \$4,600 = **\$14,700**. The higher cost in Year 2 is mainly due to the research thesis, which requires additional supervision and examination.

MBA Programme — Fee Calculation and Summary

The university adds a **20% markup** on total cost: $\$14,700 \times 20\% = \$2,940$.
Therefore: **Final MBA fee = $\$14,700 + \$2,940 = \$17,640$ per student.**

Item	Calculation	USD per Student
Direct costs — taught units (16 × \$400)	16 × \$400	6,400
Direct costs — learning materials (16 × \$75)	16 × \$75	1,200
Direct costs — assessments (16 × \$50)	16 × \$50	800
Direct costs — thesis supervision	Given	1,200
Direct costs — thesis examination	Given	500
Total direct costs		10,100
Total indirect costs		4,600
Total cost of MBA programme	$\$10,100 + \$4,600$	14,700
University markup (20%)	$\$14,700 \times 20\%$	2,940
Final fee charged	$\$14,700 + \$2,940$	17,640

MBA Programme — Cost per Academic Component

Per-Component View

Component	Number	Avg Cost (USD)
Taught units	16	750
Research thesis	1	2,700
Total	17	14,700

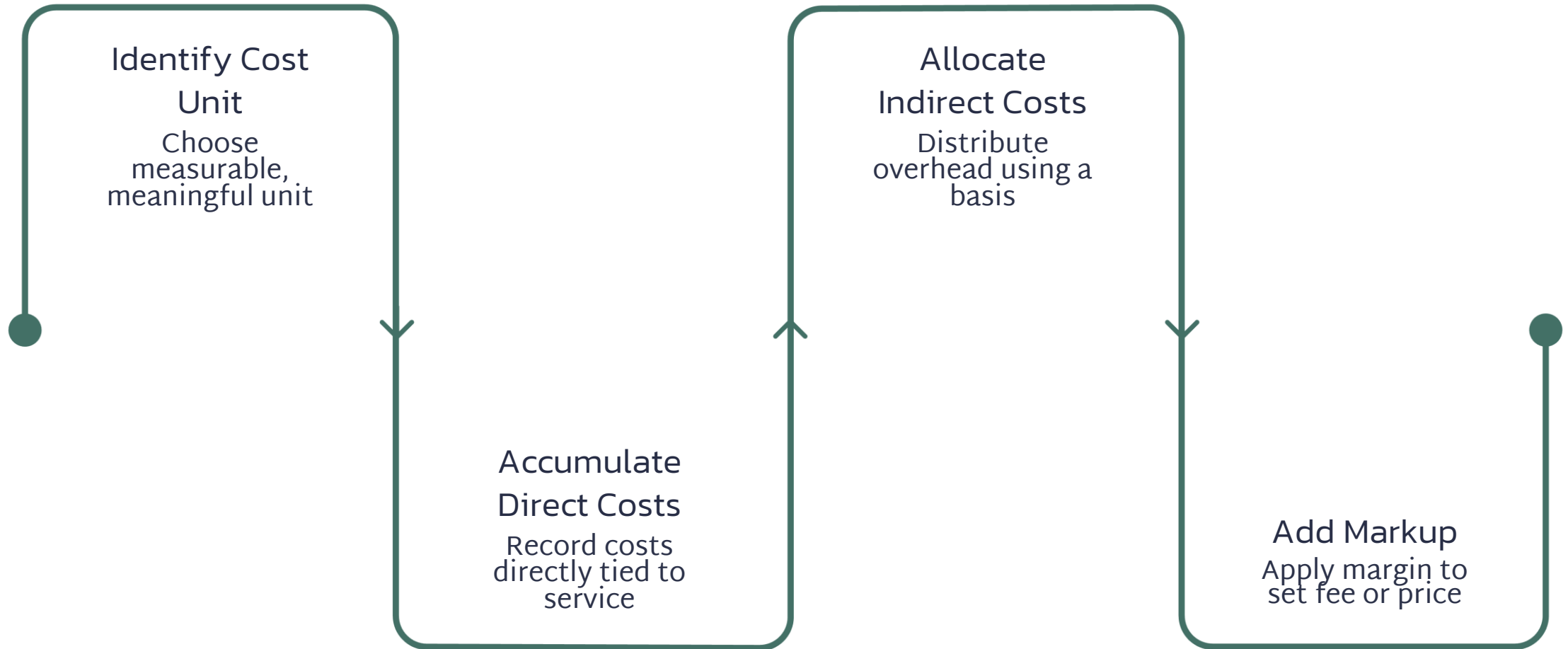
Key Point

Service costing for an MBA involves:

- Identifying direct costs attributable to teaching and research
- Allocating appropriate indirect university overheads to each student
- Determining the total cost per student
- Adding the university's required 20% return to arrive at the fee charged

The thesis is deliberately costed at a higher amount (\$2,700 vs \$750 per taught unit) because it requires more intensive academic supervision, examination and research support.

Practice Focus: Service Costing Steps



The choice of cost unit is critical it must be measurable, meaningful and reflect the nature of the service being provided. Different industries use different cost units, and the same industry may use multiple cost units depending on the decision being made.



6.7.4 Costing a Combination of Goods and Services

Some organizations offer both goods and services in one package for example, a restaurant serves (service) a meal (a good) and the two are priced as one. In this case, the **good served is usually the cost unit**. All costs are accumulated and then divided by the number of goods (meals) served.

Restaurant Example — Full Breakfast Direct Costs

A small restaurant offers a Full Breakfast (tea, coffee, pancake, egg, baked bread, fresh juice) and serves **1,000 full breakfasts per month**. A **30% markup** is applied on total cost.

Menu Item	Direct Materials (\$)	Direct Labour (\$)	Other Direct (\$)	Total Direct Cost (\$)
Tea	0.20	0.15	0.05	0.40
Coffee	0.30	0.15	0.05	0.50
Pancake	0.35	0.25	0.10	0.70
Egg	0.25	0.15	0.05	0.45
Baked bread	0.30	0.10	0.05	0.45
Fresh juice (outsourced)	0.80	—	0.05	0.85
Total	2.20	0.80	0.35	3.35

Other direct costs (\$0.35) include items specifically identifiable with a breakfast item but not direct materials or labour e.g., disposable cups, special garnish, delivery charges for outsourced juice, or special packaging for takeaway.

Outsourced fresh juice: Purchased ready-made from a supermarket. No direct labour cost for preparation. Total direct cost = \$0.80 (purchase) + \$0.05 (transport/packaging) = \$0.85.

Restaurant Example — Indirect Costs and Total Cost

Monthly Indirect Costs (allocated on total meals served)

Indirect Cost	Monthly Cost (\$)	Basis
Restaurant rent	1,500	Total meals served
Utilities — electricity, water and gas	700	Total meals served
Cleaning and sanitation	300	Total meals served
Kitchen equipment depreciation	400	Total meals served
Restaurant administration	500	Total meals served
Insurance and licences	200	Total meals served
Total indirect costs	3,600	

Indirect cost per breakfast = $\$3,600 \div 1,000 = \3.60

Total cost per full breakfast = $\$3.35$ (direct) + $\$3.60$ (indirect) = **\$6.95**

Restaurant Example — Selling Price and Summary

Applying a **30% markup** on total cost: $\$6.95 \times 30\% = \2.085 . Selling price = $\$6.95 + \$2.085 = \$9.035 \approx \mathbf{\$9.05}$

Cost Component	Computation	Cost per Breakfast (\$)
Total direct costs	$0.40+0.50+0.70+0.45+0.45+0.85$	3.35
Indirect costs	$\$3,600 \div 1,000$	3.60
Total cost	$3.35 + 3.60$	6.95
30% markup	$6.95 \times 30\%$	2.085
Selling price	$6.95 + 2.085$	\$9.05

Summary by Cost Category

Cost Category	Cost per Breakfast (\$)	% of Total
Direct materials	2.20	31.65%
Direct labour	0.80	11.51%
Other direct costs	0.35	5.04%
Indirect costs	3.60	51.80%
Total cost	6.95	100%

Recap: Seven Cost Classification Categories

1 Manufacturing vs Non-Manufacturing

Where is the cost incurred? In the factory (production) or outside (admin, selling)?

2 Product vs Period

Is the cost attached to the product (inventory) or charged to the accounting period?

3 Cost Behaviour

How does the cost change when activity changes? Variable, fixed, or semi-variable?

4 Relevant vs Irrelevant

Will the cost change because of the decision? Only relevant costs affect decisions.

5 Controllable vs Non-Controllable

Can the manager influence the cost within the relevant period and authority?

6 Avoidable vs Non-Avoidable

Can the cost be eliminated by stopping the activity or discontinuing the product?

7 Discretionary vs Non-Discretionary

Does management have discretion over the amount or timing of the cost?

Recap: Pricing Approaches



Goods Purchased for Resale

IAS 2 cost = purchase price + directly attributable costs.

Excluded: admin overheads, selling costs, unnecessary storage.

Price = Cost + Markup (or Margin).



Goods Manufactured

IAS 2 cost = direct materials + direct labour + direct expenses + production overheads (absorbed via traditional or ABC method).

Excluded: abnormal waste, admin, selling costs.



Services

Identify a cost unit (e.g., student-year, patient-day). Accumulate direct and indirect costs. Add markup or margin to determine the fee charged.



Combination (Goods + Services)

Use the good as the cost unit. Cumulate all direct and indirect costs. Apply markup to determine the menu or selling price.

Key Formulas and Relationships

Cost Function

$$Y = a + bX$$

Y = Total cost; a = Fixed cost; b = Variable cost per unit; X = Activity level

High-Low Variable Cost

$$b = (\text{High cost} - \text{Low cost}) \div (\text{High activity} - \text{Low activity})$$

Overhead Absorption Rate

$$\text{Rate} = \text{Budgeted Production Overheads} \div \text{Budgeted Absorption Base}$$

Markup Pricing

$$\text{Selling Price} = \text{Cost} \times (1 + \text{Markup \%})$$

$$\text{Markup \%} = \text{Gross Profit} \div \text{Cost} \times 100$$

Margin Pricing

$$\text{Selling Price} = \text{Cost} \div (1 - \text{Margin \%})$$

$$\text{Margin \%} = \text{Gross Profit} \div \text{Selling Price} \times 100$$

IAS 2 Inventory Cost

$$\text{Cost} = \text{Purchase price} + \text{Import duties} + \text{Freight} + \text{Handling} + \text{Non-refundable taxes} - \text{Trade discounts}$$

Topic 6 — Chapter Summary

